
E-Commerce Sales Analysis using Microsoft Excel

Process Methodology

Step 1: Understanding the Business Problem

What we did

The objective of this project was to analyze an **E-Commerce Sales Dataset** and convert raw transactional data into meaningful business insights using **Microsoft Excel**. The aim was to identify sales patterns, customer behavior, product performance, and regional trends to support business decision-making.

Business Questions

- Which age group contributes the highest sales?
 - Which gender generates the highest revenue?
 - Which product category and subcategory perform the best?
 - Which loyalty segment contributes the most revenue?
 - Which region generates the highest sales?
 - Which brands and products are top performers?
 - How do sales vary over time?
 - Do higher discounts increase sales?
 - What strategies can improve sales and profitability?
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Step 2: Understanding the Dataset

What we did

The project consists of four datasets:

- **Customer Dimension**
- **Product Dimension**
- **Store Dimension**
- **Sales Fact Table**

Each table was studied to understand:

- Primary Keys
- Foreign Keys
- Relationships
- Business meaning of each column

Customer Dimension

Contains:

- Customer ID

- Customer Age
- Gender
- Loyalty Level

Product Dimension

Contains:

- Product ID
- Product Category
- Product Subcategory
- Brand

Store Dimension

Contains:

- Store ID
- Store Name
- Region

Sales Fact Table

Contains:

- Sales ID
- Order Date
- Quantity
- Unit Price
- Discount
- Customer ID
- Product ID
- Store ID

Why?

Understanding the dataset helped establish relationships between tables before analysis.

Step 3: Data Cleaning

What we did

Before analysis, the data was cleaned to improve quality.

Activities performed:

- Checked missing values
- Removed unnecessary spaces
- Standardized text formatting
- Corrected inconsistent naming conventions
- Cleaned Loyalty Level column
- Verified consistency across tables

Formulas Used

Remove Extra Spaces

=TRIM(A2)

Convert Text to Proper Case

=PROPER(A2)

Convert Text to Upper Case

=UPPER(A2)

Convert Text to Lower Case

=LOWER(A2)

Count Blank Cells

=COUNTBLANK(A:A)

Remove Duplicate Records

Data → Remove Duplicates

Why?

Clean data improves the accuracy of calculations and analysis.

Step 4: Data Integration using VLOOKUP

What we did

Since information was distributed across multiple tables, **VLOOKUP** was used to merge required columns into the Sales Fact table.

The following information was fetched:

- Customer Details
- Product Details
- Category
- Subcategory
- Brand
- Store Name
- Region
- Loyalty Level

Example Formula

=VLOOKUP(Customer_ID, Customer_Table, 2, FALSE)

Product Category

=VLOOKUP(Product_ID, Product_Table, 3, FALSE)

Product Subcategory

=VLOOKUP(Product_ID, Product_Table, 4, FALSE)

Brand

=VLOOKUP(Product_ID, Product_Table, 5, FALSE)

Store Name

=VLOOKUP(Store_ID,Store_Table,2,FALSE)

Region

=VLOOKUP(Store_ID,Store_Table,3,FALSE)

Loyalty Level

=VLOOKUP(Customer_ID,Customer_Table,5,FALSE)

Why?

This created a **single consolidated Sales Analysis table**, making further analysis easier.

Step 5: Handling Missing Values & Calculated Fields

Some important fields contained blank values. Instead of deleting records, missing values were calculated using mathematical relationships.

A. Calculating Total Amount

Formula

=(Unit_Price*Quantity)*(1-Discount)

Mathematical Formula

Total Amount = (Unit Price × Quantity) × (1 – Discount)

Why?

Calculates the final amount paid after discount.

B. Calculating Unit Price

When Unit Price was blank:

Formula

=Total_Amount/(Quantity*(1-Discount))

Mathematical Formula

Unit Price = Total Amount ÷ [Quantity × (1 – Discount)]

Why?

Determines the original price per unit.

C. Calculating Discount Percentage

When Discount was blank:

Formula

$=1-(\text{Total_Amount}/(\text{Unit_Price}*\text{Quantity}))$

Mathematical Formula

$\text{Discount} = 1 - (\text{Total Amount} \div (\text{Unit Price} \times \text{Quantity}))$

Why?

Calculates the percentage discount applied.

D. Calculating Quantity

When Quantity was blank:

Formula

$=\text{Total_Amount}/(\text{Unit_Price}*(1-\text{Discount}))$

Mathematical Formula

$\text{Quantity} = \text{Total Amount} \div [\text{Unit Price} \times (1 - \text{Discount})]$

Why?

Calculates the number of units purchased.

Step 6: Data Formatting

What we did

Formatted data for consistency:

- Unit Price → Currency
- Total Amount → Currency
- Discount → Percentage
- Order Date → Date Format

Why?

Improves readability and reporting accuracy.

Step 7: Creating Derived Columns

Age Range

Customers were grouped into:

- 18–25
- 26–35
- 36–45
- 46–55

- 56–65

Formula

```
=IF(B2<=25,"18-25",  
IF(B2<=35,"26-35",  
IF(B2<=45,"36-45",  
IF(B2<=55,"46-55","56-65"))))
```

Why?

To perform age-wise sales analysis.

Month

Formula

```
=TEXT(Order_Date,"MMM")
```

Example:

15-03-2025 → Mar

Year

Formula

```
=YEAR(Order_Date)
```

Example:

15-03-2025 → 2025

Year-Month

Formula

```
=TEXT(Order_Date,"yyyy-mm")
```

Example:

15-03-2025 → 2025-Mar

Why?

To analyze monthly and yearly sales trends.

Step 8: Statistical Analysis

Statistical measures were calculated for:

- Total Amount
- Unit Price
- Discount
- Customer Age
- Quantity

Measure	Formula
Mean	=AVERAGE(range)
Median	=MEDIAN(range)
Mode	=MODE.SNGL(range)
Minimum	=MIN(range)
Maximum	=MAX(range)
Count	=COUNT(range)
Variance (Sample)	=VAR.S(range)
Variance (Population)	=VAR.P(range)
Standard Deviation (Sample)	=STDEV.S(range)
Standard Deviation (Population)	=STDEV.P(range)
Skewness	=SKEW(range)
Kurtosis	=KURT(range)

Why?

To understand:

- Central tendency
- Data variability
- Distribution
- Customer purchasing behavior

Step 9: Creating Pivot Tables

Pivot Tables were created to summarize the dataset and answer business questions.

Age-wise Sales Analysis

Analyzed:

- Count of Sales
- Sum of Total Amount

Gender-wise Sales Analysis

Compared:

- Male
- Female

based on:

- Orders
- Revenue

Category vs Subcategory Analysis

Analyzed:

- Beauty
- Clothing
- Electronics
- Home
- Sports

along with their subcategories.

Loyalty Level Analysis

Compared:

- Bronze
- Silver
- Gold
- Platinum

customers.

Region-wise Sales Analysis

Compared:

- North
 - South
 - East
 - West
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Top Brands Analysis

Ranked brands based on total sales.

Top Products Analysis

Ranked products based on revenue.

Monthly & Yearly Trend Analysis

Analyzed sales across different Year-Month combinations.

Discount vs Sales Analysis

Compared:

- Discount Percentage
- Sales Performance

to understand the effectiveness of discounts.

Why?

Pivot Tables summarize large datasets into meaningful business information.

Step 10: Creating Pivot Charts

After creating Pivot Tables, Pivot Charts were developed to visualize the analysis.

Charts created include:

- Column Chart → Age-wise Sales
- Pie Chart → Gender-wise Sales
- Line Chart → Monthly & Yearly Trend
- Pie Chart -> Subcategory Vs Sales

Charts make trends and comparisons easier to understand and communicate.

Step 11: Dashboard Creation

An interactive dashboard was built using:

- KPI Cards
- Pivot Tables
- Pivot Charts
- Slicers

KPI Cards Created

- Total Sales
- Total Orders

Slicers Added

- Sub Category
- Region
- Loyalty Level
- Year
- Payment type

Why?

The dashboard allows users to dynamically filter data and quickly analyze business performance.

Step 12: Business Analysis

Using the dashboard, four levels of analytics were performed.

Descriptive Analysis

Question Answered: *What happened?*

Summarized sales performance, customer demographics, category performance, and regional contributions.

Diagnostic Analysis

Question Answered: *Why did it happen?*

Explained the reasons behind high-performing categories, customer segments, and regional differences.

Predictive Analysis

Question Answered: *What is likely to happen?*

Forecasted future trends based on historical sales patterns.

Prescriptive Analysis

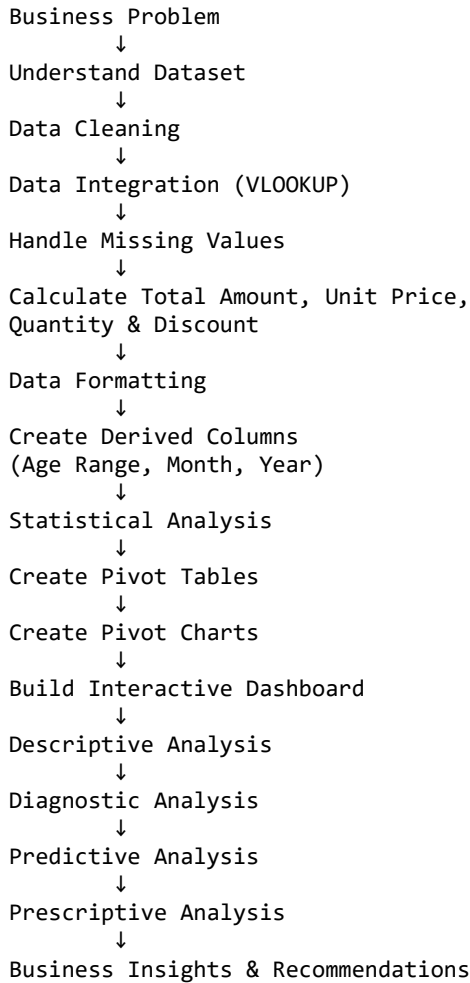
Question Answered: *What should the business do?*

Provided recommendations such as:

- Invest more in Sports and Electronics.
- Improve the Beauty category.
- Expand business in the West region.
- Strengthen customer loyalty programs.
- Increase average order value through cross-selling and bundling.
- Optimize discount strategies.

- Use personalized marketing campaigns.
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Overall Process Flow



This consolidated methodology provides a complete, step-by-step explanation of **what was done, why it was done, the formulas used, and how the project progressed from raw data to an interactive business dashboard and actionable insights.**